Status	Finished
Started	Saturday, 24 October 2026, 11:21 PM
Completed	Saturday, 24 October 2026, 11:25 PM
Duration	3 mins 46 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It complicates the analysis process
b. It makes the analysis slower
c. It has no effect on the analysis process
d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **2**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Dividing the input into smaller subproblems
b. Sorting the entire list sequentially
c. Combining sorted sublists
d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort
b. Selection sort
c. Quick sort
d. Merge sort

The correct answer is:
Merge sort

Question **4**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are combined to form a single sorted list
b. They are compared element by element
c. They are discarded
d. They are split again into smaller sublists

The correct answer is: They are combined to form a single sorted list

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Merge Sort
b. Linear Search
c. Binary Search
d. Bubble Sort

The correct answer is: Bubble Sort

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Bubble Sort
b. Quick Sort
c. Merge Sort
d. Insertion Sort

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the largest element to its correct position
b. Bubbling up the smallest element
c. Sorting the entire list in one pass
d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **8**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It requires external libraries
b. It sorts data out of the box efficiently
c. It is less efficient than custom sorting algorithms
d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently

Question **9**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It does not require recursion
b. It sorts data using a single pass
c. It is based on the divide-and-conquer approach
d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **10**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Dynamic programming
b. Greedy method
c. Backtracking
d. Divide and conquer

The correct answer is:
Divide and conquer

Question **11**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. sort()
b. order()
c. arrange()
d. sorted()

The correct answer is: sort()

Question **12**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Merging
b. Rearranging
c. Sorting
d. Searching

The correct answer is: Sorting

Question **13**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It complicates the selection of items
b. It slows down the process
c. It makes it easier to select items based on their relationship to the rest of the items
d. It makes the data unsorted

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **14**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Distribution
b. Extraction
c. Selection
d. Insertion

The correct answer is:
Insertion

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Bubble Sort
- b. Heap Sort
- c. Merge Sort
- d. Quick Sort

The correct answer is: Bubble Sort